No. : 02

Date :

PRACTICAL 1

Title -
Import the legacy from different sources such as (Excel, Text CSV, OData) and load in the target system

Theory -
Data collection is the important component of business intelligence. Data is collected from the various source in the various format. The selection process of the data includes data currency, data quality the level of detail in the data. BI Architecture may require structured / unstructured data.

Solution -

① Importing data from Excel in PowerBI

step 1: Start PowerBI desktop

step 2: Click on GetData on the Home Ribbon and select Excel

step 3: Select any Excel sheet which you want

step 4: In Navigator Pane, you will get below screen, Check the sheet and data of the sheet will be displayed in the Navigation Pane

step 5: If you want to edit the sheet you can click on Transform Data

step 6: After that you can visualize your data in many field like bar chart, graph, table format, etc

② Importing data from text/csv in Power BI

step 1 : Start Power BI desktop

step 2 : Click on Get data on the Home Ribbon and Select Text/csv

step 3 : Select any Text/csv sheet which you want

step 4 : In Navigator Pane, you will get the below screen, check the sheet and the data of sheet will be displayed in the Navigation Pane

step 5 : If you want to edit the sheet you can click on Transform data

step 6 : After that you can visualized your data in many field like bar chart, graph, table format, etc

③ Importing data from OData in Power BI

step 1: Start Power BI desktop

step 2: Click on Get Data on Home Ribbon and Select OData feed

step 3: In the OData feed Select Basic option and give http://services.odata.org/v3/Northwind/Northwind.svc and Click OK button

step 4: In the Navigator Pane, you will get the list of the table of the Northwind OData feed. Select Orders table after the content of the table is loaded in the preview. Click Transform data to edit and query on the table

step 5: After that you can visualize your data in many field like bar chart, graph, table format, etc.

No. :

Date :

PRACTICAL 2

Title :
Sales, Profit and Loss analysis using Power BI

Aim:
To analyze company sales data and calculate profit and loss using Power BI for better business analysis

Theory:
Power BI is a data visualization and business intelligence tool used to analyze data from Excel. It helps in calculating sales, profit and loss using simple formulas and display results through charts for easy understanding

Solution:

Step 1 - Open Power BI desktop

Step 2 - Load Excel Data
Go to Home → Get Data → Excel, select Company Excel file, choose sheet 1 and click Load

Step 3 - Verify Data Columns
Check that following columns are available
- Product
- Selling Price
- Cost Price
- Year

Step 4 - Create Profile / Loss Column
Click Modeling → New Column and other
Profit_Loss = 'sheet 1'[Selling Price] - 'sheel 1'[cost Price]

Step 5 - Create Total Sales Measure
Go to Modeling → New Measure and enter
Total Sales = SUM('sheet 1'[Selling Price])
then Add Total Sales Card Visual

Step 6 : Create Product - Wise Profit / Loss Chart

Select a clustered column chart

- X axis → Product
- Y c Value → Profit_Loss

Step 7 : Format the Visuals & Analyze the Results

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PRACTICAL 3

Topic -
Implementation of classification Algorithm R Programming

Solution -

```
>rainfall.timeseries<-c(799,1174.8,865.1,1334.0,0.6,635.4,
                        918.5,685.5,998.6,784.2,985,882.8,
                        1071)
>print(rainfall.timeseries)
 [1] 799.0,1174.8,865.1,1334.0,0.6,635.4,918.5,685.5,998.6,
     784.2,985,882.8,1071

>rainfall.timeseries<-ts(rainfall,start=c(2012,1),
                         frequency=12)
>print(rainfall.timeseries)
        Jan    Feb    Mar    Apr    May    Jun
2012  799.0  1174.8  865.1  1334.0   0.6   635.4

        Jul    Aug    Sep    Oct    Nov    Dec
        918.5  685.5  998.6  784.2  985.0  882.8

2013    1071.0

>C<-2
>png(file="rainfall.png")
>dev.off()
null device
          1

>plot(rainfall.timeseries)
```

PRACTICAL 4

Title-
Practical implementation of Decision Tree using R Toot

Theory-
Decision Tree is a graph to represent choices and their results in the form of a tree. The nodes in graph represent an event or choice and the edges of the graph represent the decisions rules or conditions.

Solution -
Executes the following command in R

```
> install.packages ("Party")
  Warning in install.package ("party"):
    'lib ="C:/Program Files /R/R-3.5.2/library" is not
    writable
 -- Please select CRNA Mirror for use in table session --
also installing the dependencies 'TH.data','multcamp',
 'mutnorm','modeltools' ,'strucchange', 'coin', 'zoo','sandwich'

>library (Party)
  Loading required package : grid
  Loading required package : mutnorm
  Loading required package : modeltools
  Loading required package : stats4
  Loading required package : structure
  Loading required package : zoo

  Attaching package : 'zoo'

  The following objects are masked from 'Packageibase';

        as.Date, as.Date.numeric
  Loading required package : sandwich
```

```
>print(head(readingSkills))
  nativeSpeaker age shoesize score
1           Yes   5 24.83189 32.29385
2           Yes   6 25.95238 36.63105
3            No  11 30.42170 49.60593
4           Yes   7 28.66450 40.28456
5           Yes  11 31.88207 55.46085
6           Yes  10 30.07843 52.83124

>input.dat <- readingskills[c(1:105),]
>input.dat
   nativespeaker age shoesize score
1            Yes   5 24.83189 32.29385
2            Yes   6 25.95238 36.63105
3             No  11 30.42170 49.60593
4            Yes   7 28.66450 40.28456
5            Yes  11 31.88207 55.46085
6            Yes  10 30.07843 52.83124
7             No   7 27.25963 34.40229
8            Yes  11 30.07843 52.83124
9            Yes   5 27.25963 34.40229
10            No   7 26.69835 33.93269
11           Yes  11 31.86645 55.46876
12           Yes  10 29.15575 51.34140
13            No   9 26.85613 41.77098
14            No   6 24.23420 30.03304
15            No   5 24.23420 25.62268
16           Yes   6 25.67538 35.30042
17            No   5 24.86357 25.62843
18            No   6 26.15357 30.76591

>png(file="decision_tree.png")
>output.tree <- (tree(nativespeaker ~ age + shoesize + score,
                      data = input.dat)
>dev.off()
  null device
            1
```

No. : 10

Date :

PRACTICAL 5

Title -

Practical implementation of k-means clustering using R tools

Theory -

k-means clustering intends to partition to n objects into k clusters in which each object belongs to the clusters with nearest mean. This method product exactly k different clusters of greatest possible number or distinction. The best number of clusters k leading to the greatest separation

Solution -

Execute following command in R

```
>iris
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	Setosa
5	5.0	3.6	1.4	0.4	setosa
6	5.4	3.9	1.7	0.3	Setosa
7	4.6	3.4	1.4	0.2	Setosa
8	5.0	3.4	1.5	0.2	Setosa
9	4.4	2.9	1.4	0.1	Setosa
10	5.0	3.1	1.5	0.2	setosa
11	4.4	3.7	1.6	0.2	Setosa
12	4.8	3.4	1.5	0.2	Setosa

```
>newiris <- iris
>newiris
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	~~Species~~
1	5.1	3.5	1.4	0.2	"
2	4.9	3.0	1.4	0.2	"
3	4.7	3.2	1.3	0.2	"
4	4.6	3.1	1.5	0.2	"
5	5.0	3.6	1.4	0.2	"

6	5.4	3.9	1.7	0.4	"
7	4.6	3.4	1.4	0.3	"

```
>newirisspecies <- NULL
>newiris
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
6	5.4	3.9	1.7	0.4
7	5.6	3.4	1.4	0.3
8	4.0	3.4	1.5	0.2
9	4.4	2.9	1.4	0.2
10	4.9	3.1	1.5	0.1

```
>(kc<-(kmeans(newiris,3)))
```

k-means clustering with 3 clusters of sizes 50, 62, 38

cluster means -

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.006000	3.428000	1.462000	0.246000
2	5.901613	2.748387	4.393548	1.433871
3	6.850000	3.073684	5.742105	2.071053

Clustering vector:

```
[1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2
[52] 2 2 2 2 2 2 2 2 2 2 2 3 2 3 3 3 3 3 2 3 2 3 2 3 2 3 2 3 2 3 2
```

Within cluster sum of squares by cluster:

```
[1] 15.13100 39.82097 23.87947
```

(between_SS / total_SS = 88.4%)

Available Components:

```
[1] "cluster"   "centers"   "totss"     "withiness" "size"
    "tot.withiness" "between"   "iter"    "ifault"
```

```
> table (iris$species, KE$cluster)
```

	1	2	3
setosa	50	0	0
versicolor	0	48	2
virginica	0	14	36

```
> plot (newiris [c("Sepal.Length", "Sepal.Width")], col = KS$cluster)

> points (KC$centers[, c("Sepal.Length", "Sepal.Width")], col = 1:3,
        pch = 8, cex = 2)
```

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